

1. Explain the concept of **API (Application Programming Interface)**, why do we need **APIs**.
 2. Compare **developer API** vs **application API (normal APIs)**
 3. Name some different types of **APIs**
 4. Compare **path variables** vs **request parameters** in **REST API**.
 5. Explain the different components that make up a **RESTful API** and what does each part do?
 6. Explain what **cURL** is and why we use **API testing tools** like **Postman** instead of testing **APIs** directly with **cURL**.
 7. List common **HTTP status codes** and their meanings.
 8. List **HTTP methods** and their meanings, and their expected **HTTP status codes**.
 9. Explain why **REST API** is **stateless**.
 10. Discuss about best practices for **REST API design**, from **performance** perspective.
 11. Explain the concept of **XSS (Cross-Site Scripting)** and **CSRF (Cross Site Request Forgery)** and how to avoid them.
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API Practices:

Use **Postman** or other **API testing tools** to:

1. Find at least 5 different **public APIs** (e.g., **GitHub APIs**, **Google Cloud APIs**, **GeoInfo APIs**, **Weather APIs**) and use them to explain what defines a **REST API**. These **APIs** can use any **HTTP methods** and may also include **non-REST APIs** (e.g., **GraphQL**). Some **public APIs** may require **API keys** (user registration required);
2. Justify whether these **APIs** follow **API design best practices**, and provide your better design for them.
3. List the above **APIs** in form of **cURL** commands , and attach **Postman** screenshots in your markdown submission.
4. List the **request headers** and **response headers** of the **APIs** mentioned above, and explain what each **key-value pair** in the **headers** section does.